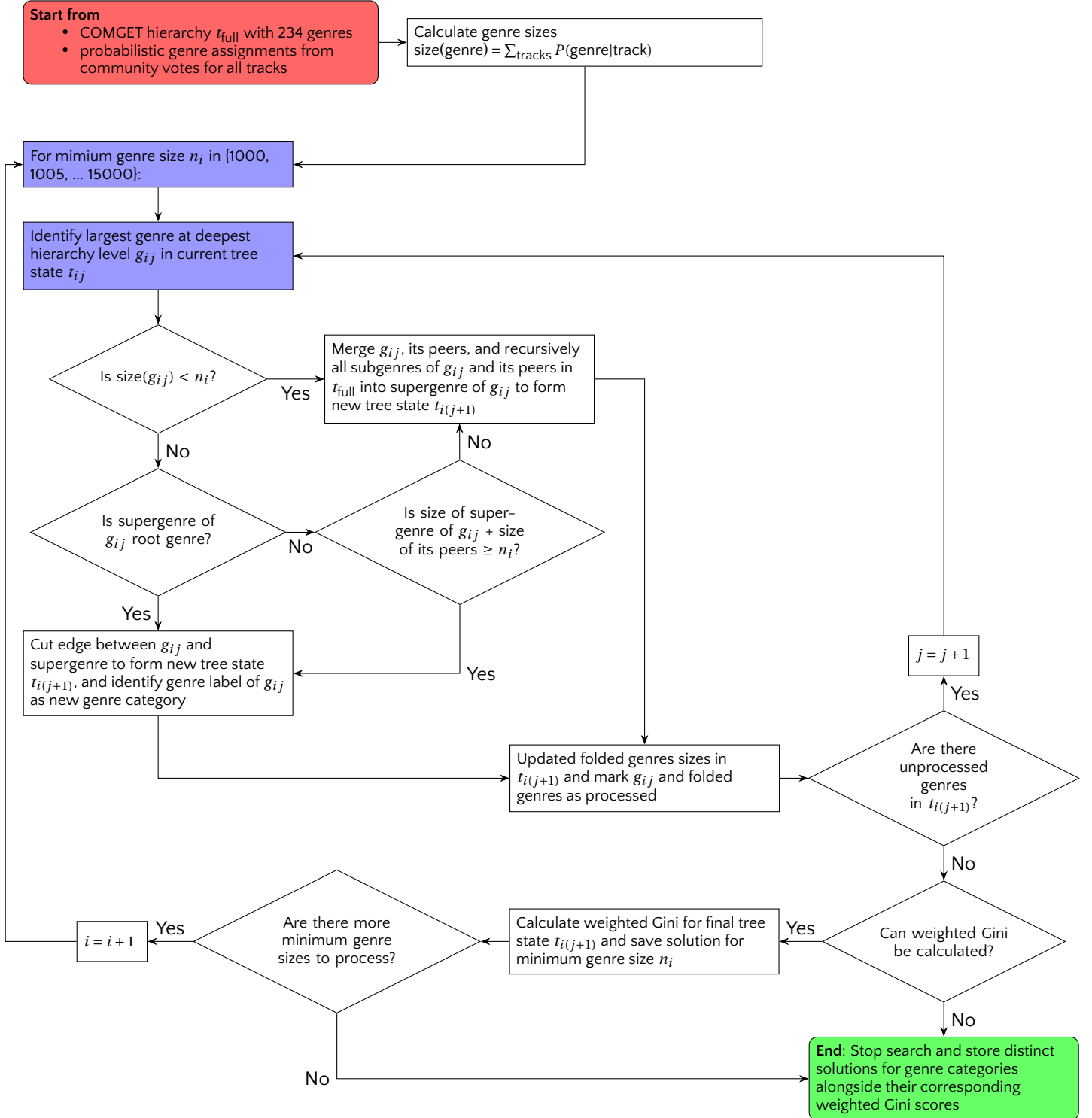


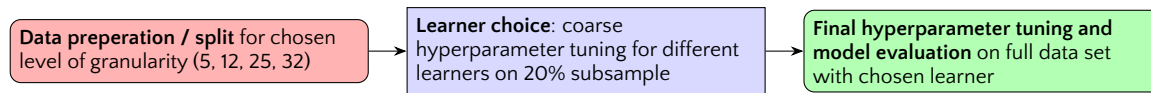
Appendix

A. Detailed FOLDCAT flowchart for folding COMGET to identify genre categories

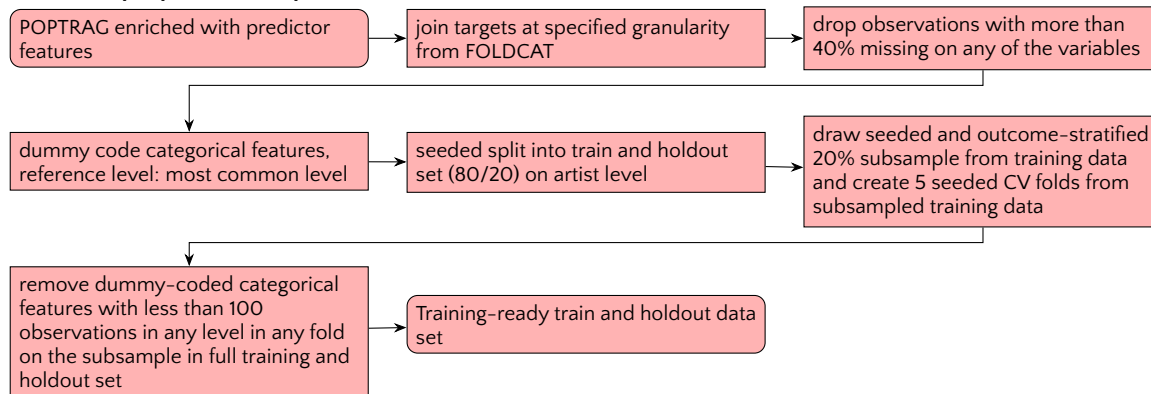


B. RQ3 Machine Learning Workflow

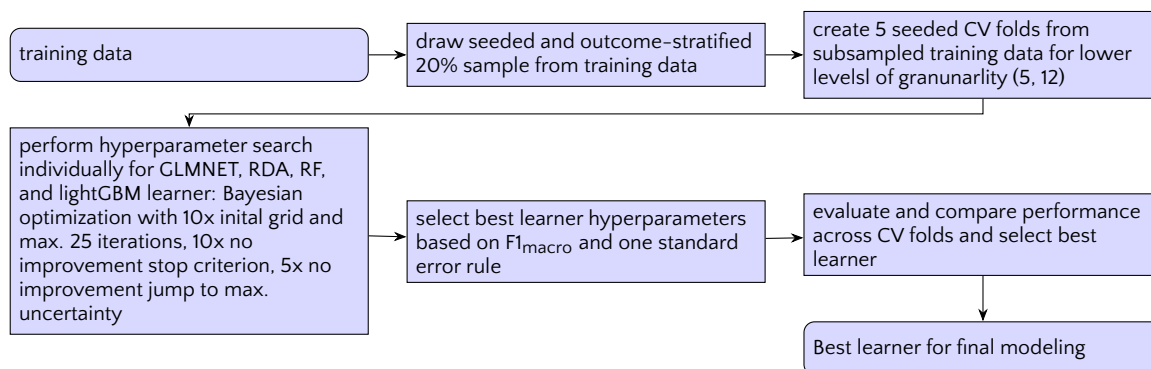
Overall Workflow



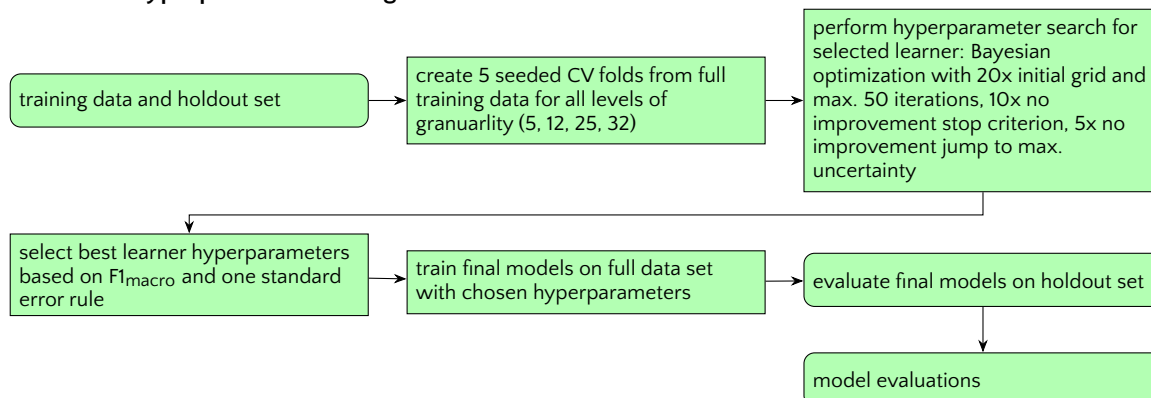
B.1 Data preparation / split



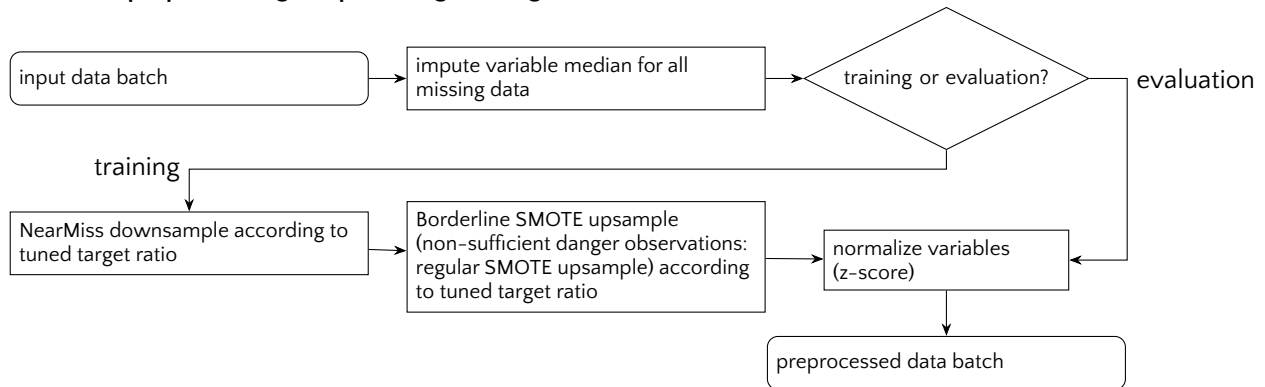
B.2 Learner Choice



B.3 Final Hyperparameter tuning and model evaluation



B.4 Data preprocessing recipe during training and evaluation



C. RQ3 features

Construct	Subdimensions	Data Type	Indicators	Variable Content	Source
genre of a popular music track	release milieu (via metadata)	metadata release	artist	gender persona country of activity country of origin birth year is dead? 27 Spotify distribution genres	MusicBrainz
			release	release year indie label? D-A-CH label? release length	Spotify
		metadata lyrics	code system	language track explicit?	MusicBrainz Spotify
			information content	distinct words ratio repeated lines ratio	Genius
	musical style (via content analysis)	lexial content analysis of lyrics	emotion	positive negative anger disgust fear joy sadness suprise anticipation trust	Genius + syuzhet
			topics	sentiment	Genius + sentimentr
		machine-learning-based audio signal analysis	sound of production	loudness brightness acousticness liveness typicality genre sounds	Spotify (Echo-Nest) Acousticbrainz (Essentia)
			instrumnetation	instrumentalness tonality voice and female voice sound electronic sound	
			composition	tempo chords structure duration time signature key rhythm danceability emotional expression mood	
	popularity (via sucess metrics)	streams, followers, and release numbers on commercial music streaming platforms	track	popularity rank	Spotify Deezer
			artist	streams followers number of releases	Spotify Deezer
			label	median track populariy median release popularity median track popularity	Spotify

Table C.1: Features used for training the automated classifier and their theoretical dimension, adapted from anonymous authors (XXXX).

D. RQ3 classification hyperparameters and outcomes for higher granularities

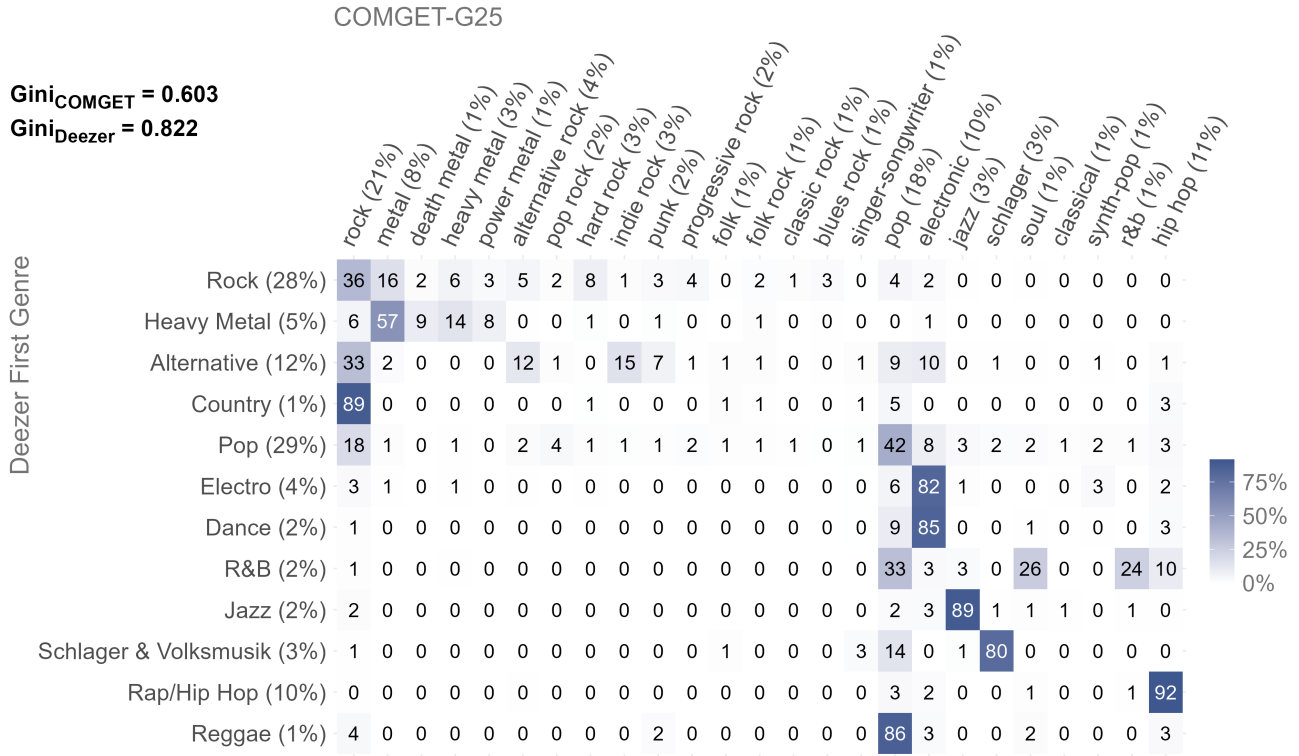


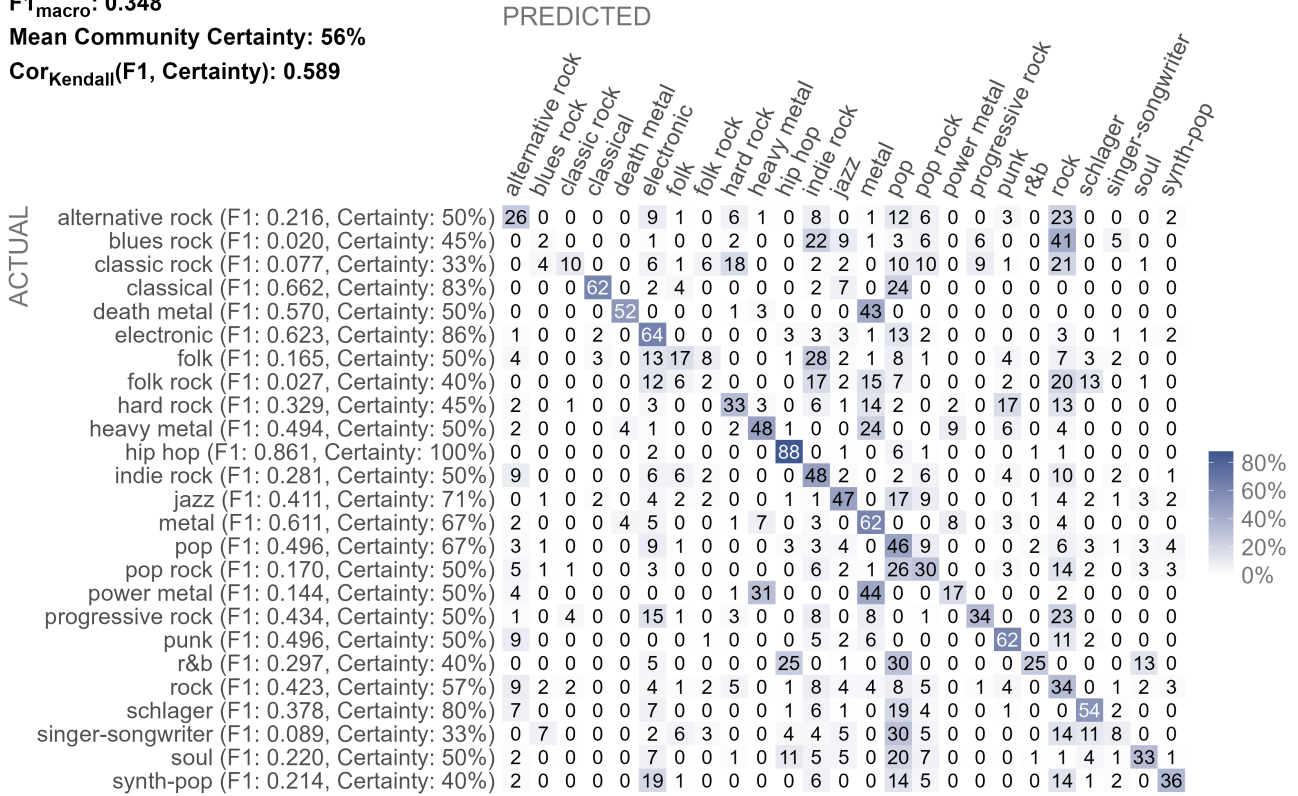
Figure D.1: Contingency tables comparing COMGET-G25 to release genres from the *Deezer* API; 27 genres were found in the data set; values show $P(\text{COMGET} | \text{External})$. Discarded 15 genres for visualization with less than 0.5% relative frequency: *African Music*, *Asian Music*, *Bolero*, *Brasilian Music*, *Classical*, *Disco*, *Film / Video Games*, *Folk*, *German Music*, *Kids*, *Latin Music*, *Ranchera*, *Singer & Songwriter*, *Soul & Funk*, *Turkish Folk Music*.

# genres	boosting rounds [50, 2000]	tree depth [3, 15]	feature fraction [1, # of predictors]	leaf size [1, 100]	learning rate [0.001, 0.1]	target ratio [1, max/min]
5	93	12	68	46	0.072	4.815
12	1145	10	5	28	0.046	2.116
25	1512	9	85	91	0.005	2.641
32	176	9	55	15	0.0182	4.589

Table D.1: Hyperparameter settings for final LightGBM models at different granularity levels.

$F1_{\text{macro}}$: 0.348

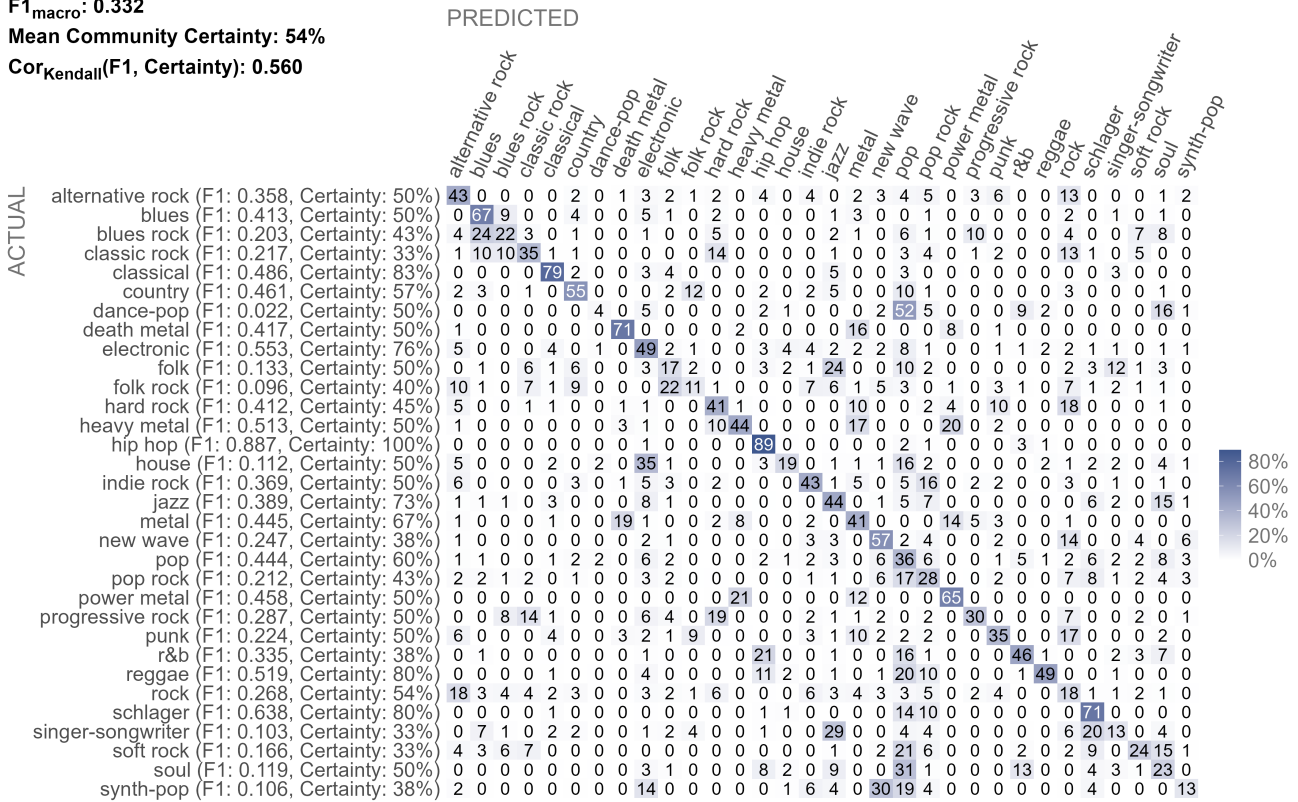
Mean Community Certainty: 56%

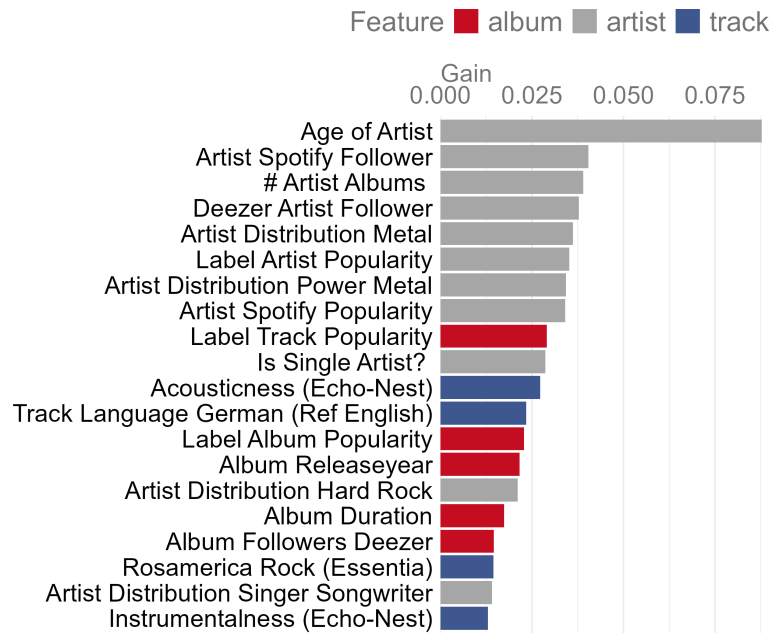
 $\text{Cor}_{\text{Kendall}}(F1, \text{Certainty})$: 0.589

(a) COMGET-G25

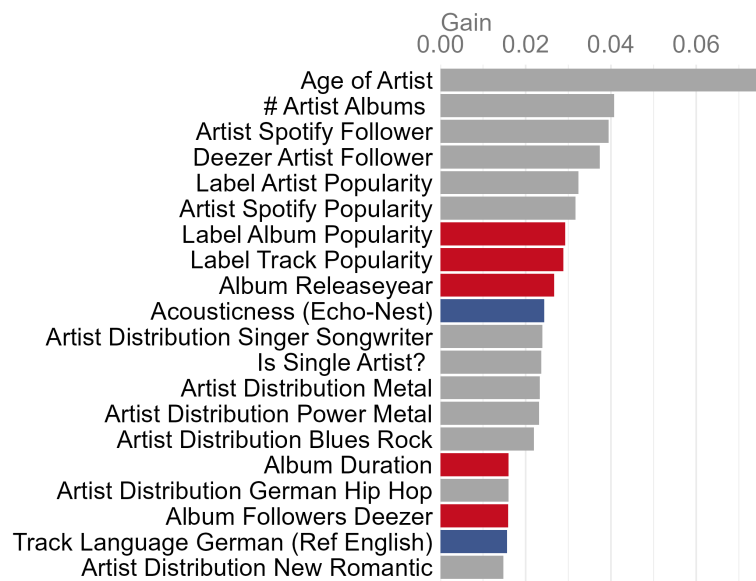
 $F1_{\text{macro}}$: 0.332

Mean Community Certainty: 54%

 $\text{Cor}_{\text{Kendall}}(F1, \text{Certainty})$: 0.560



(a) COMGET-G25



(b) COMGET-G32

Figure D.3: Top 20 most important features for automated classification at high and very high granularity. Values show relative feature importance from the final LightGBM models.

E. Github Repository

Please find further details and the complete code for reproduction hosted on *Github*:

Link is left out for the purpose of anonymization of authors

Data is available from the authors on request.